

Chapter 4: Use of Multimedia

Multimedia is the integration of multiple forms of media, such as text, graphics, audio, video, and animation, to convey information in an interactive and engaging manner.

4.1 Digital Graphics

Graphics are a core component of multimedia. They are categorized into two main types:

- **Raster (Bitmap) Graphics:** Composed of tiny dots called **pixels**. When enlarged, they may lose quality (pixelation). Common formats: JPEG, PNG, GIF, BMP.
- **Vector Graphics:** Created using mathematical formulas and geometric objects (points, lines, curves). They can be scaled to any size without losing quality. Common formats: SVG, AI, EPS.

Key Concepts in Graphics

- **Resolution:** The number of pixels in an image, usually measured in PPI (Pixels Per Inch). Higher resolution means better quality.
- **Color Depth:** The number of bits used to represent the color of a single pixel.
- **Compression:** Reducing file size. *Lossy* compression loses some data (e.g., JPEG), while *Lossless* retains all data (e.g., PNG).

4.2 Animation

Animation is the process of creating the illusion of motion by displaying a sequence of static images (frames) rapidly.

- **2D Animation:** Created in a two-dimensional space (e.g., traditional cartoons).
- **3D Animation:** Created in a three-dimensional environment with depth, often used in modern movies and games.
- **Frame Rate:** The number of frames displayed per second (FPS). Standard animation often uses 24 or 30 FPS.

4.3 Audio and Video

Audio: Digital sound representation. Formats include MP3 (compressed), WAV (uncompressed), and MIDI (musical instructions).

Video: A sequence of images combined with audio. Formats include MP4, AVI, and MOV. Video quality depends on resolution (e.g., 1080p, 4K) and bit rate.

4.4 Software Tools

Common multimedia editing software mentioned in the syllabus:

Type	Software Examples
Graphic Editing	GIMP, Adobe Photoshop, Inkscape
2D/3D Animation	Synfig Studio, Blender, Adobe Animate
Audio Editing	Audacity
Video Editing	OpenShot, Adobe Premiere Pro

4.5 Multimedia Distribution

Multimedia can be distributed through various platforms including web pages, mobile apps, television, and optical discs (DVD/Blue-ray). Factors like file size and bandwidth are critical for online distribution.